

MODULE 3

Factors affecting Ground Water Recharge

Groundwater recharge is the process by which water moves from the surface into the groundwater system. Several factors can influence the rate and extent of groundwater recharge. These include:

1. **Precipitation:** The amount, intensity, and frequency of rainfall play a crucial role in recharging groundwater. Higher and more consistent rainfall leads to greater recharge, while prolonged dry periods may reduce recharge rates.
2. **Soil Type and Permeability:** The texture and structure of the soil affect how easily water can infiltrate and recharge groundwater. Sandy soils tend to have higher permeability, allowing water to pass through more easily, while clayey or compacted soils can impede water infiltration.
3. **Land Use and Vegetation Cover:** Areas with dense vegetation or forest cover typically experience higher recharge rates due to improved infiltration and reduced surface runoff. Urbanization and deforestation can reduce the natural recharge process by increasing runoff and reducing soil permeability.
4. **Topography:** The landscape's shape can influence how water moves across the ground. Steep slopes may cause water to run off quickly, reducing the time available for infiltration. In contrast, flatter terrain allows water to infiltrate the ground more efficiently.
5. **Human Activities:** Over-extraction of groundwater, improper land management, and activities like agriculture, construction, or mining can reduce recharge rates. For example, over-extraction can create a negative balance in the groundwater system, leading to depletion.
6. **Climate:** Seasonal variations and long-term climate changes (like droughts or floods) significantly affect recharge. Warmer climates may increase evaporation rates, reducing the amount of water available for infiltration, while colder climates can result in snowmelt that contributes to recharge during spring.

7. **Geology and Rock Types:** The composition and structure of the underlying geology can affect recharge. For example, areas with fractured or porous rocks (such as limestone or sandstone) allow water to move more freely into the aquifer, leading to more recharge.
8. **Water Table Depth:** If the water table is very deep, recharge may take longer or be less efficient due to the distance water has to travel. Shallower water tables are more easily recharged.
9. **Surface Water Availability:** Proximity to rivers, lakes, or other bodies of water can influence groundwater recharge, as surface water can seep into the ground and replenish aquifers.

These factors can interact in complex ways, and local conditions often determine the overall effectiveness of groundwater recharge in a specific region.

Reviving traditional techniques for Water Harvesting

Reviving traditional water harvesting techniques is gaining attention worldwide due to its potential to address modern water scarcity challenges in a sustainable manner. These age-old methods, developed by various cultures over centuries, often use locally available materials and are designed to optimize water use in regions with irregular or limited rainfall. Here are some key notes on this revival:

1. Importance of Traditional Water Harvesting
 - **Sustainability:** These methods are often low-cost, low-tech, and require minimal maintenance, making them more sustainable than modern, large-scale systems.
 - **Climate Adaptation:** Traditional techniques are well adapted to local climates and can be resilient against extreme weather events, such as droughts or floods.
 - **Community Empowerment:** Many traditional methods rely on community participation and can be a way to build social cohesion around shared water resources.
2. Key Traditional Water Harvesting Techniques
 - **Step Wells (Baolis in India):** These are deep wells with a series of steps leading down to the water table. They were used in regions with high temperatures and irregular rainfall to provide a steady water

supply for drinking, irrigation, and daily use.

- **Rainwater Harvesting Pits (Cisterns):** Common in many parts of the world, these underground or surface structures collect rainwater from roofs or other surfaces for later use. In some African regions, such as the Sahel, traditional cisterns are still in use.
- **Water Channels (Zing or Qanat):** Developed in ancient Persia, qanats are underground channels that tap into aquifers to transport water over long distances, reducing evaporation loss in hot climates.
- **Terracing and Contour Farming:** In areas prone to soil erosion, such as in the Andes or Southeast Asia, terraces are built into hillsides to slow water flow and increase infiltration, improving soil moisture for crops.
- **Percolation Tanks (India):** These are small, localized reservoirs constructed to capture rainwater and allow it to percolate into the ground, replenishing groundwater reserves.
- **Water Harvesting from Roofs (Gulams in the Middle East):** Rainwater collected from rooftops, especially in arid regions like the Middle East, was often stored in large cisterns for household use.

3. Benefits of Reviving Traditional Techniques

- **Cost-Effective:** Traditional methods are often cheaper to build and maintain than modern infrastructure.
- **Eco-Friendly:** These techniques use natural processes and locally sourced materials, with minimal environmental impact.
- **Improved Water Availability:** In regions suffering from erratic rainfall, these methods can ensure a reliable supply of water.
- **Conserving Groundwater:** Many traditional methods, such as percolation tanks and rainwater harvesting, help recharge groundwater aquifers.

4. Challenges in Reviving Traditional Techniques

- **Loss of Knowledge:** Traditional techniques are often not passed down due to urbanization and modernization. Reviving these methods requires rediscovering and preserving local knowledge.
- **Climate Change:** While these methods are resilient, they may not be enough to cope with increasingly unpredictable weather patterns or larger-scale droughts caused by climate change.
- **Integration with Modern Systems:** Combining traditional methods with contemporary water management technologies can be a ^{Help} challenge. Effective integration requires understanding both systems'

strengths and limitations.

5. Revival Strategies

- **Education and Awareness:** Educating communities about the benefits of traditional water harvesting techniques is crucial to their revival. Workshops, demonstrations, and local government support can help.
- **Government Policies and Incentives:** Encouraging the adoption of traditional techniques through subsidies, incentives, and policy support can accelerate their revival.
- **Community-Led Initiatives:** Involving local communities in designing, building, and maintaining water harvesting systems ensures that they meet local needs and are sustainable.
- **Research and Documentation:** Recording and studying traditional techniques can provide valuable insights into how they can be adapted and applied in contemporary settings.

6. Case Studies

- **India:** The revival of step wells, particularly in drought-prone areas of Rajasthan, has helped improve water accessibility. Programs to restore traditional ponds and tanks have seen success in some regions.
- **Mexico:** The use of rainwater harvesting through traditional systems like *aljibes* (cisterns) has been promoted in rural areas, especially in regions with limited access to piped water.
- **Morocco:** The use of *Khattaras* (underground irrigation systems) to tap into groundwater has been revived in some arid areas to support agriculture.

In conclusion, reviving traditional water harvesting techniques offers a promising approach to address the growing water scarcity problem. With the right integration of modern technology, these methods could be a sustainable and effective solution, especially for rural and remote areas where resources are limited.

Calculation of available rain water for harvesting

Calculating the amount of rainwater available for harvesting involves considering various factors such as the area of the catchment, the intensity of rainfall, and the efficiency of the system. Here's a step-by-step guide to help you estimate the available rainwater for harvesting:

1. Catchment Area:

- The catchment area is the surface where rainwater is collected, usually the roof. The larger the catchment area, the more rainwater can be harvested.
- **Formula for Area:** $\text{Area} = \text{Length} \times \text{Width}$
 $\text{Area} = \text{Length} \times \text{Width}$
- Measure the roof or surface area in square meters (m^2) where rainwater will be collected.

2. Rainfall (Precipitation):

- Rainfall data is typically obtained from meteorological reports or local rainfall statistics. The rainfall is measured in millimeters (mm).
- For instance, 1 mm of rainfall on 1 m^2 of surface gives 1 liter of water.

3. Rainwater Harvesting Efficiency:

- No system is 100% efficient. Factors such as evaporation, overflow, and filtration reduce the total volume of rainwater that can be stored.
- Efficiency typically ranges between 70% and 90%, depending on the system design.
- You can adjust efficiency by using the formula:
 $\text{Effective Rainwater} = \text{Rainfall} \times \text{Catchment Area} \times \text{Efficiency Factor}$
- For example, with a 0.8 efficiency factor (80% efficiency), the rainwater harvested would be reduced by 20%.

4. Calculating the Volume of Harvested Water:

- Once you have the catchment area and rainfall data, use the following formula to calculate the total volume of rainwater that can be harvested:

$$\text{Volume} = \text{Catchment Area} \times \text{Rainfall} \times \text{Efficiency Factor}$$

- Example: If you have a roof area of 100 m^2 , rainfall of 50 mm, and efficiency of 80%:
 $\text{Volume} = 100\text{m}^2 \times 50\text{mm} \times 0.8 = 4000\text{liters}$
- Note: 1 mm of rainfall on 1 m^2 = 1 liter of water, so convert the rainfall depth (mm) into volume by multiplying by the area.

5. Storage and Overflow Considerations:

- Ensure that your storage system (e.g., rainwater tank) has adequate capacity for the volume of water you intend to capture.
- Overflow systems should be planned to redirect excess water safely during heavy rain periods.

6. Seasonal and Geographic Variations:

- Rainfall varies by season and location, so it's important to factor in seasonal fluctuations when planning for year-round harvesting.

Example Calculation:

Let's assume:

- Catchment Area: 150 m^2 (roof area)
- Rainfall: 100 mm (10 cm)

- Efficiency Factor: 0.85 (85%)

Using the formula:

$$\text{Volume} = 150\text{m}^2 \times 100\text{mm} \times 0.85$$

$$\text{Volume} = 150 \times 100 \times 0.85 = 12,750\text{liters}$$

So, 12,750 liters of rainwater would be available for harvesting under these conditions.

PREPARATION OF SUITABLE TECHNICAL DRAWING AND DESIGN OF RAINWATER HARVESTING STRUCTURE

When preparing a technical drawing and design for a rainwater harvesting structure, the goal is to ensure efficiency, functionality, and sustainability. Here are some key points to consider:

1. Site Analysis

- **Topography and Slope:** Assess the site's slope to determine the gravity flow of water. The system should use the natural slope for efficient water flow.
- **Roof Area:** The size of the roof will determine the water harvesting capacity. Identify which areas of the roof will be used for collecting rainwater.
- **Soil Type:** Assess soil permeability if considering underground storage. Clay-rich soils may require different considerations compared to sandy soils.

2. Design of the Collection System

- **Roof Design:** The material and pitch of the roof influence water runoff. A sloped roof is ideal for directing water to the catchment system. The roof should be free of debris.
- **Gutters and Downspouts:** Proper placement and sizing of gutters and downspouts are essential for efficient water collection. They should be designed to handle maximum rainfall during storms. Downspouts should lead to the storage tank or filtration system.
- **First Flush System:** Incorporate a first flush diverter to ensure that the initial dirty water from the roof is discarded before reaching the storage tank.

3. Storage System

- **Tank Size:** The size of the storage tank should be based on the roof area and local rainfall patterns. Typically, the size is calculated by multiplying the roof area by the average annual rainfall to estimate the volume of water that can be harvested.
- **Material:** Tanks can be made from materials such as concrete, plastic, or metal. The choice of material depends on durability, cost, and the local environment.
- **Tank Location:** Position the tank in a place where it is easily accessible for maintenance and cleaning. It should also be close enough to the downspout for

easy connection but not in a location that may interfere with other site features.

- **Overflow System:** Design an overflow system to divert excess water away from the tank and prevent flooding or tank damage.

4. Filtration and Purification

- **Pre-Filtration:** Install mesh filters at the top of the downspouts to prevent large debris from entering the storage system.
- **Post-Filtration:** A finer filtration system, such as a sand filter, can be used before the water enters the tank to remove smaller particles.
- **Purification:** Depending on the intended use of the water (drinking, irrigation, etc.), additional purification methods like UV treatment or chlorination may be required.

5. Pump and Distribution System

- **Pump Design:** If the harvested rainwater is to be used for household or irrigation purposes, design an appropriate pump system to deliver water from the tank to the desired locations. Ensure the pump is appropriately sized for the expected demand.
- **Plumbing Layout:** Plan the distribution system carefully, ensuring there are no leaks, and consider water-saving features like pressure regulators or drip irrigation.

6. Water Use and Maintenance

- **Usage Plan:** Determine the intended use of the harvested rainwater (e.g., irrigation, toilet flushing, landscape watering). This will affect the design and filtration needs.
- **Maintenance Access:** Include access points in the design for cleaning filters, inspecting the tank, and performing necessary repairs.
- **Monitoring:** Consider adding water level indicators or gauges to monitor the water stored and avoid overflow.

7. Regulations and Standards

- **Local Regulations:** Check local laws and regulations for rainwater harvesting, as some areas have specific codes or requirements regarding water quality, plumbing, or tank installation.
- **Design Standards:** Follow any relevant national or international standards for rainwater harvesting system design, such as those for plumbing, construction, and water treatment.

8. Technical Drawing

- **Dimensions and Scale:** Draw detailed, scaled plans for the entire system, including roof layout, gutter and downspout positions, tank size, pump placement, and filtration units.
- **Elevation Views:** Provide elevation views to show the vertical relationships between components like gutters, storage tanks, and pumps.
- **System Diagram:** Include a schematic diagram of the system that shows the flow of water from collection to storage and distribution.

- **Details:** Include construction details, such as tank foundations, pipe sizes, material specifications, and fittings.
- **Structural Considerations:** If the tank is elevated, include structural designs to support the weight and pressure of the water.

Example Components of a Technical Drawing:

- **Plan View:** Shows the layout of gutters, downspouts, tank location, and pipe connections.
- **Section View:** Depicts the vertical alignment of elements like roof, gutters, downspouts, filters, and tanks.
- **Isometric View:** For a 3D understanding of the system's layout and connections.

By following these points, the rainwater harvesting structure can be designed to operate effectively, be durable, and provide a sustainable water source. Proper planning and technical drawings help ensure that all components are well-integrated and functional.

Provide a numerical example for calculating the annual rainwater harvested, we would need some specific data. Let's walk through a typical example:

Given:

1. **Roof Area (Catchment Area)** = 100 m²
2. **Average Annual Rainfall** = 800 mm (0.8 meters)
3. **Runoff Coefficient (C)** = 0.8 (since some water is lost due to evaporation or absorption by the roof)
4. **Efficiency of the system** (let's assume) = 85% (some water might be lost during the collection and storage process)

Formula to Calculate Rainwater Harvested:

Rainwater Harvested = Catchment Area × Rainfall × Runoff Coefficient × Efficiency

Now, let's plug the numbers into the formula:

Rainwater Harvested = 100m² × 0.8m × 0.8 × 0.85

Rainwater Harvested = 100 × 0.8 × 0.8 × 0.85 = 54.4m³

Result:

So, the total amount of rainwater harvested in a year is **54.4 cubic meters**.

Design a rainwater harvesting system for a small residential house with the following details:

Roof Area: 150 m² Average Annual Rainfall: 1000 mm,

Runoff Coefficient: 0.8

Storage Tank Capacity: The tank should store rainwater to meet the household's needs for 3 months. The daily water requirement for the household is 500 liters. Design total volume of rainwater harvested per year, Size of the storage tank, Overflow and filtration system

Solution:

To design a rainwater harvesting system for the small residential house with the provided details, we can break it down into several key steps.

1. Determine the Total Volume of Rainwater Harvested per Year

The total volume of rainwater harvested depends on the roof area, the average annual rainfall, and the runoff coefficient.

Formula:

Total Rainwater Harvested per Year (m³) = Roof Area × Rainfall × Runoff Coefficient

Where:

- Roof Area = 150 m²
- Rainfall = 1000 mm = 1 meter (since 1000 mm = 1 m)
- Runoff Coefficient = 0.8

So:

Total Rainwater Harvested per Year = 150 × 1 × 0.8 = 120 m³

Thus, 120 cubic meters (120,000 liters) of rainwater can be harvested per year.

2. Determine the Size of the Storage Tank

The storage tank should store enough water to meet the household's needs for 3 months. The daily water requirement for the household is 500 liters.

To calculate the total water needed for 3 months:

Total Water Needed for 3 Months = 500 × 90 = 45,000 liters

Thus, the storage tank should have a capacity of 45,000 liters to meet the household's needs for 3 months.

3. Overflow System

The overflow system ensures that excess rainwater is safely redirected when the storage tank is full. The overflow should be designed to handle the maximum rainfall during a storm event.

Typically, the overflow is connected to the nearest stormwater drainage system or a designated drainage area. The sizing of the overflow pipe should be based on local rainfall intensity, but a standard approach is to use an overflow pipe with a diameter

that can handle at least the expected peak flow rate.

For simplicity, let's assume the overflow pipe can handle all excess water from the roof during heavy rainfall.

4. Filtration System

A filtration system is essential to ensure that the water collected is clean and safe to use. The typical filtration system includes:

- **First flush diverter:** This prevents the initial dirty runoff (which may contain debris, dust, and other pollutants) from entering the storage tank.
- **Pre-storage filtration:** A fine mesh filter to capture larger debris (e.g., leaves and twigs).
- **Post-storage filtration:** A finer filtration system (such as a UV filter or activated carbon filter) for drinking and other potable uses.

A basic filtration system would look like this:

1. **First Flush Diverter:** Directs the first 10-15% of rainfall away from the tank.
2. **Mesh Filter:** Installed on the downspout, typically with a 0.5-1 mm mesh size to capture larger particles.
3. **Storage Tank Filtration:** A fine mesh filter at the inlet of the tank, possibly combined with UV treatment or activated carbon if the water will be used for drinking.

Final System Summary:

- **Total Volume of Rainwater Harvested per Year:** 120,000 liters.
- **Storage Tank Capacity:** 45,000 liters (to cover 3 months of household water usage).
- **Overflow System:** Designed to handle excess water when the tank is full (with appropriate piping).
- **Filtration System:**
 - First flush diverter to divert the initial dirty water.
 - Mesh filter to capture larger debris.
 - Post-storage filtration (e.g., UV or activated carbon filter) for potable water use.

This system will help provide a reliable source of rainwater for the household while ensuring the water is clean and safe to use.

NUMERICAL

Designing a rainwater harvesting (RWH) system involves calculating the amount of water that can be collected from a given catchment area and ensuring the system is sized appropriately for storage and use. Here's a breakdown of how to approach numericals for RWH design:

Given Data:

1. **Catchment Area:** The area from which rainwater is collected (in square

meters, m^2).

2. **Rainfall:** The average annual rainfall in the region (in millimeters, mm).
3. **Runoff Coefficient:** A factor that accounts for the efficiency of the catchment surface in collecting water (this depends on the surface type, for example, a roof might have a runoff coefficient of 0.8).
4. **Storage Capacity:** The amount of water to be stored (in liters or cubic meters).

Step 1: Determine the Total Rainfall Volume

Total rainfall volume VVV (in cubic meters) can be calculated using the formula:

$$V = \text{Catchment Area} \times \text{Rainfall Depth} \times \text{Runoff Coefficient}$$

Where:

- Catchment Area is in square meters (m^2).
- Rainfall Depth is in millimeters (mm), so convert it to meters by dividing by 1000.
- Runoff Coefficient is typically a value between 0.5 and 0.9 depending on the surface material (e.g., 0.8 for concrete, 0.7 for tiled roofs, etc.).

Step 2: Convert the Volume to Liters (If Needed)

Since 1 cubic meter (m^3) = 1000 liters (L), multiply the result by 1000 to get the volume in liters.

Example 1: Calculation for a House Roof

Given:

- Catchment Area = 100 m^2
- Annual Rainfall = 1200 mm
- Runoff Coefficient = 0.8

Solution:

1. Convert rainfall from mm to meters: $1200\text{mm}=1.2\text{m}$
2. Calculate the total volume of rainwater collected: $V=100\text{m}^2 \times 1.2\text{m} \times 0.8$
 $V=96\text{m}^3$
3. Convert the volume to liters: $V=96\text{m}^3 \times 1000=96,000\text{L}$

So, the total volume of rainwater collected in a year would be **96,000 liters**.

Example 2: Sizing the Storage Tank

If the system needs to store 2 months' worth of water usage for a family:

Given:

- Monthly water usage: 10,000 liters
- Rainwater harvesting contribution: 2 months of supply

Solution:

1. Calculate the required storage capacity:
Storage Capacity = $10,000\text{L/month} \times 2\text{months} = 20,000\text{L}$

So, a storage tank with a **capacity of 20,000 liters** would be required to meet the 2-month water needs of the family.

These are some basic examples, but depending on your local conditions and the specific use case (e.g., irrigation, drinking water), you may need to factor in additional details such as filtration, overflow systems, and seasonal variations.